

Fitness & Wellness Mobile App for Member Self-Service

Implemented as "WellFit" — Fitness & Wellness Platform

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Problem Statement

- Manual processes and long waits for class & trainer bookings
- Class and trainer booking handled inconsistently
- Data entry errors and difficulty accessing member fitness records
- Poor coordination between membership, plans, and progress tracking
- No centralized way to monitor member activity and engagement
- Result: fitness centers need one connected self-service platform, not scattered tools

Objectives

- Centralize fitness center operations on a single mobile platform
- Simplify member registration and class/trainer booking
- Give members personalized workout tracking and exercise plans
- Reduce booking conflicts through class & trainer scheduling
- Streamline membership, wellness plans, and progress monitoring
- Reduce paperwork and minimise manual data errors

Proposed System

- One centralized mobile platform for all fitness center operations
- Members register, book classes/trainers, and track progress online
- Trainers manage client plans and their session schedules
- Class & trainer booking reduces scheduling conflicts
- Membership, wellness plans, and progress data stay linked to member profiles
- Admin dashboard gives staff a centralized view of activity

System Architecture

Layered Architecture Diagram



Modules & Functionalities



Member Registration



Class & Trainer Booking



Trainer Schedule Mgmt



Workout & Fitness Tracking



Class Booking & Waitlist



Membership & Billing



Wellness & Nutrition Plans



Progress & Body Metrics



Admin Dashboard

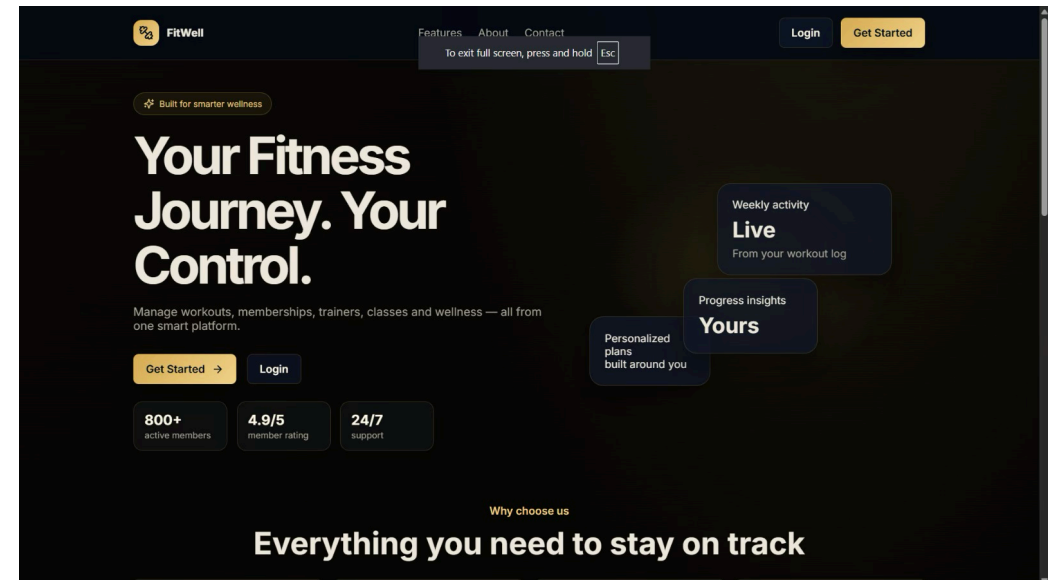


Notifications (SMS)

Frontend Design

- Three role-based portals: Admin, Trainer, Member
- Each portal opens to a personalized Dashboard
- Sidebar navigation for Bookings, Workouts, Membership, Plans, Progress
- Forms for class booking, membership & trainer session notes
- Built with React.js using Vite for fast development

Development Environment

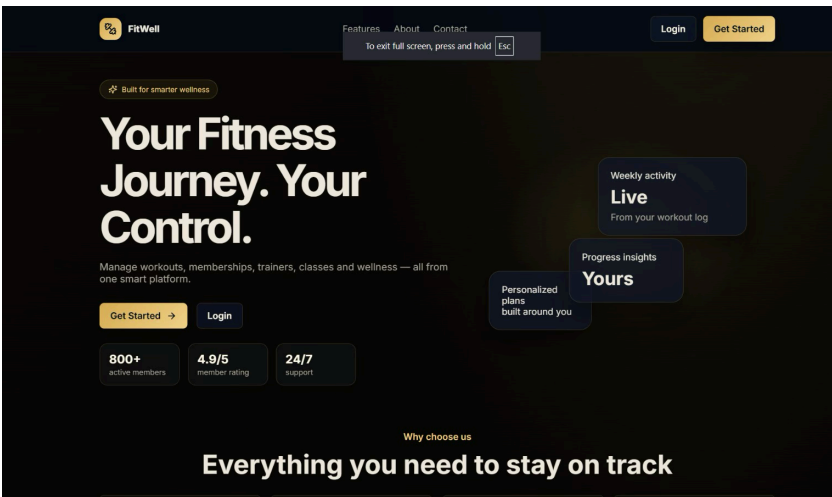


React + Vite dev server running WellFit locally

Frontend Implementation — Screenshots

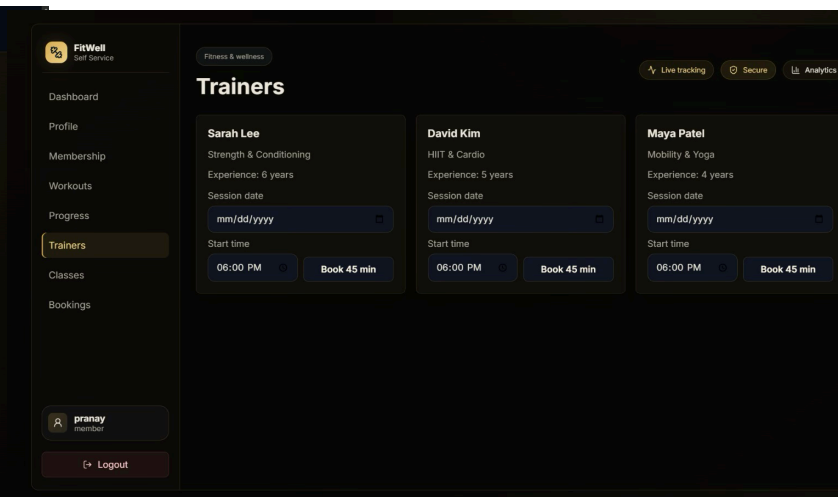
Live dashboards for each user role, built and running in WellFit

Admin Portal



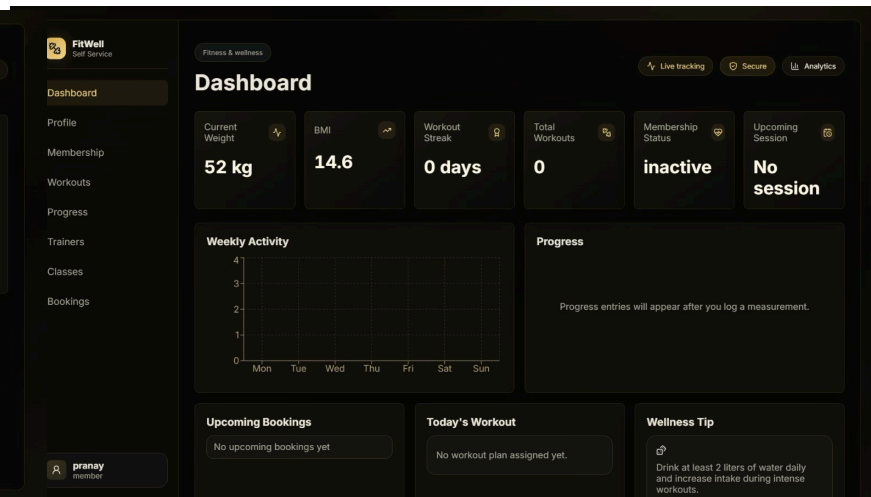
Live stats, schedule & booking overview

Trainer Portal



Today's sessions & client consultations

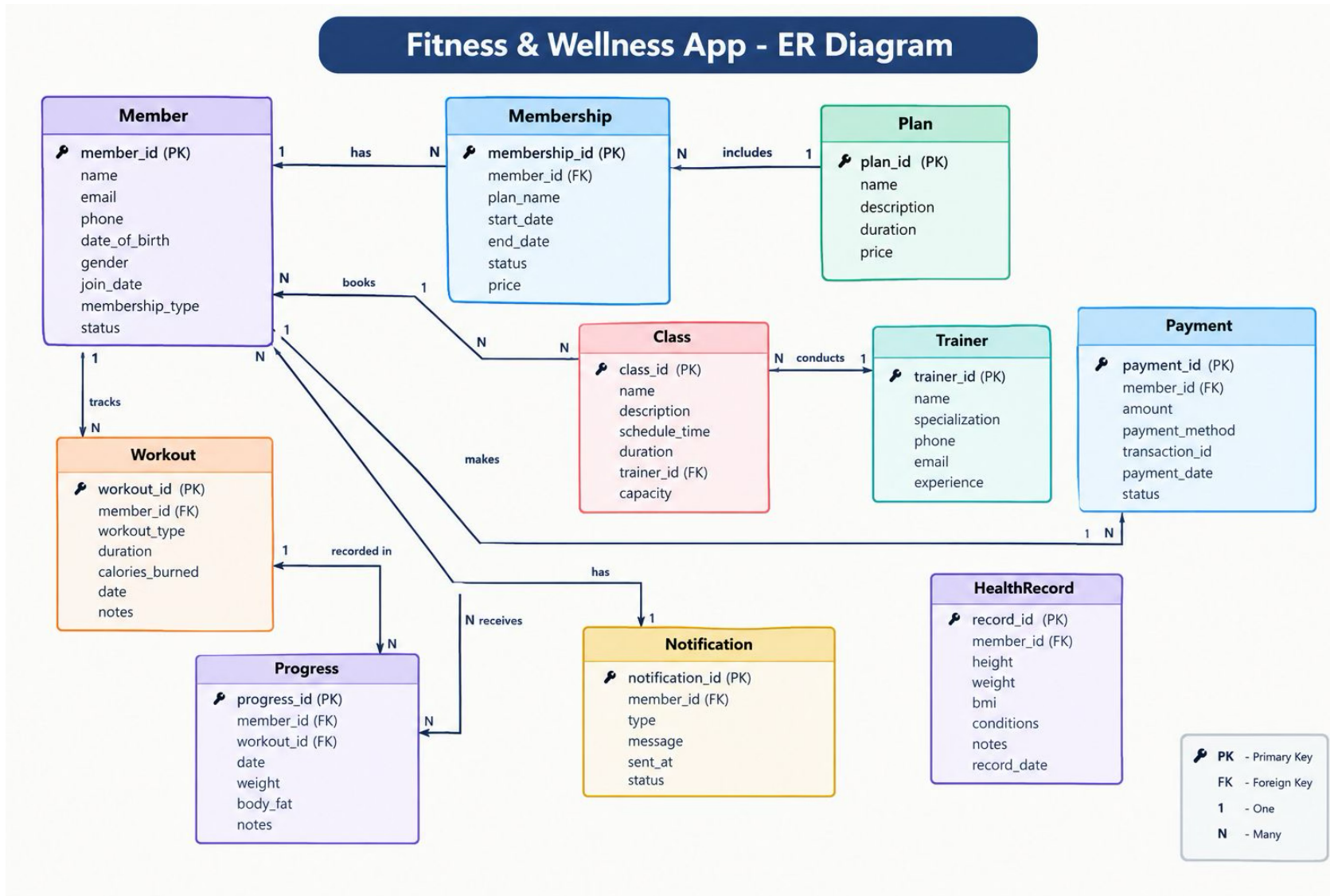
Member Portal



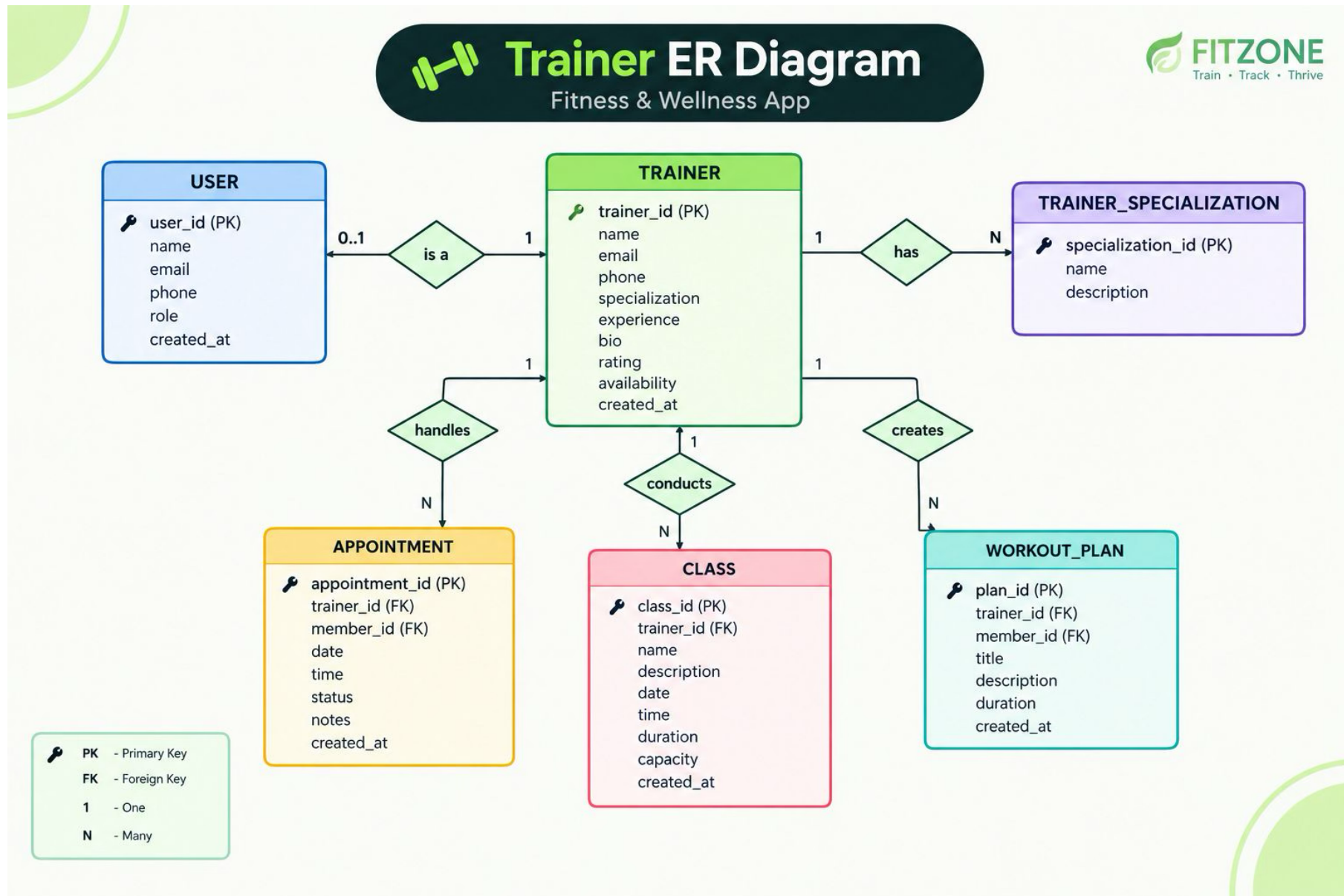
Membership, payments & plan overview

All three portals share consistent WellFit branding, navigation, and card-based UI styling.

Database Design — ER Diagram



Database Design — ER Diagram



Database Schema & Sample Records

Views Implemented

- member_bookings — class & trainer bookings
- trainer_schedule_view — trainer availability
- workout_plan_view — member workout data
- membership_details — membership & billing
- progress_report_view — member progress & metrics
- Transactions verified: INSERT → UPDATE → COMMIT

Current Progress

- Frontend built with React.js (Vite) — Admin, Trainer & Member portals live
- MySQL database designed and implemented (wellfit_db schema)
- Core views built for bookings, membership, plans & progress records
- Sample data inserted and tested with transactional updates
- Backend REST API integration (Node.js/Express, JWT) in progress
- UI styled and functional across all three role-based dashboards

Challenges / Issues Faced

- Structuring a relational schema that keeps membership, billing & plans consistent
- Keeping class/trainer booking status synced across portals
- Implementing consistent role-based access across three dashboards
- Designing a UI simple enough for members, detailed enough for trainers & admins

Future Work

- Complete backend REST API integration with JWT authentication
- Add SMS notifications via Twilio
- Build an analytics dashboard for member engagement insights
- Explore AI-based personalized workout recommendations
- Deploy using Docker for production readiness

Conclusion

- Centralizes fitness & wellness operations on a single, connected platform
- Frontend (3 role-based portals) and database implementation show real progress
- Remaining work: backend API integration, notifications, and deployment

Thank You